

Homological Algebra

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Chapter 1

List 1

Problem 1. State the definition of monomorphism, and prove that an injective function of sets is a monomorphism in $\mathbf{Set}$.

Solution: A monomorphism $f \in \text{Hom}_{\mathcal{C}}(B, C)$ is a morphism that is left cancelative in the sense that for any object A and morphisms $g, h \in \text{Hom}_{\mathcal{C}}(A, B)$, we have:

$$fg = fh \Rightarrow g = h. \quad (1.1)$$

Now, if $\mathcal{C} = \mathbf{Set}$, the equality $fg = fh$ implies $f(g(a)) = f(h(a))$, $\forall a \in A$ and if f injective, by definition, we have $g(a) = h(a)$, $\forall a \in A$, i.e., $g = h$.

Problem 2. Prove that $h : \mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism in $\mathbf{Ring}$.

Solution: Let $f, g : \mathbb{Q} \rightarrow R$ be two ring morphisms such that $fh = gh$, i.e., $f(n) = g(n)$, $\forall n \in \mathbb{Z}$. Here, we assume that a morphism in $\mathbf{Ring}$ send unity to unity, so $f(1) = g(1) = 1_R$. Furthermore, since $\mathbb{Q}$ is a field, $f : \mathbb{Q} \hookrightarrow R$ and $f(\mathbb{Q}^\times) \subset R^\times$. In particular $m1_R \in R^\times$. Now we have

$$m1_R(f(\frac{1}{m}) - g(\frac{1}{m})) = 0 \Rightarrow f(\frac{1}{m}) = g(\frac{1}{m}). \quad (1.2)$$

This proves that $f = g$.

Problem 3. Write the proof that any two terminal objects in a category $\mathcal{C}$ are isomorphic to each other.

Solution: Let T, T' be two terminal objects in $\mathcal{C}$. Let $f \in \text{Hom}_{\mathcal{C}}(T, T')$ and $T' \in \text{Hom}_{\mathcal{C}}(T', T)$, then $f \circ g \in \text{Hom}_{\mathcal{C}}(T, T) = \{1_T\}$. Therefore $fg = 1_T$ and in a similar way we see that $gf = 1_{T'}$. This proves that f is an isomorphism.

Problem 4. State an "explicit" definition of the free product $G * H$ of groups G and H . Prove that the free product is a coproduct in Grp .

Solution: Let $A = G \amalg H$, we define

$$G * H = \coprod_{n \geq 1} A^n / \sim \quad (1.3)$$

where the equivalence relation is defined by

$$\begin{aligned} (a_1, \dots, a_i, a_{i+1}, \dots, a_n) &\sim (a_1, \dots, a_i a_{i+1}, \dots, a_n) \text{ if } a_1, a_2 \in G \text{ or } a_1, a_2 \in H \\ (a_1, \dots, a_{i-1}, a_i, a_{i+1}, \dots, a_n) &\sim (a_1, \dots, a_{i-1}, a_{i+1}, \dots, a_n) \text{ if } a_i = 1_G \text{ or } a_i = 1_H \end{aligned} \quad (1.4)$$

The multiplication is defined by

$$[(a_1, \dots, a_n)] * [(b_1, \dots, b_m)] = [(a_1, \dots, a_n, b_1, \dots, b_m)]. \quad (1.5)$$

By the equivalence relations defined above, the multiplication rule is clearly well defined.

We see that $(G * H, *)$ is a group. The associativity and existence of unity is clear and the inverse operation is defined by

$$[(a_1, a_2, \dots, a_n)]^{-1} = [(a_n^{-1}, \dots, a_2^{-1}, a_1^{-1})]. \quad (1.6)$$

Now we define morphisms

$$i_1 : G \rightarrow G * H : g \mapsto [(g)], \quad i_2 : H \rightarrow G * H : h \mapsto [(h)]. \quad (1.7)$$

We claim that $(G * H, i_1, i_2)$ is a coproduct of G and H . Suppose that we have morphisms $f_1 : G \rightarrow K, f_2 : H \rightarrow K$, then we define a morphism

$$f : G * H \rightarrow K : [(a_1, \dots, a_n)] \mapsto f_{j_1}(a_1) \cdots f_{j_n}(a_n) \quad (1.8)$$

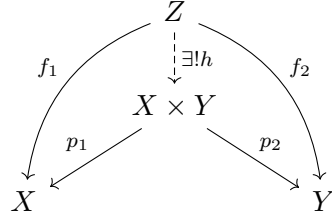
where we let $j_k = 1$ (resp. 2) if $a_k \in G$ (resp. H). Furthermore, we see that $f i_j = f_j$, $j = 1, 2$. For uniqueness, we observe that if $h : G * H \rightarrow K$ is a morphism, such that $h i_j = f_j$, $j = 1, 2$, then $h([(g)]) = f_1(g)$ and $h([(h)]) = f_2(h)$. So

$$h([(a_1, \dots, a_n)]) = h([(a_1)] * \cdots * [(a_n)]) = h([(a_1)]) \cdots h([(a_n)]) \quad (1.9)$$

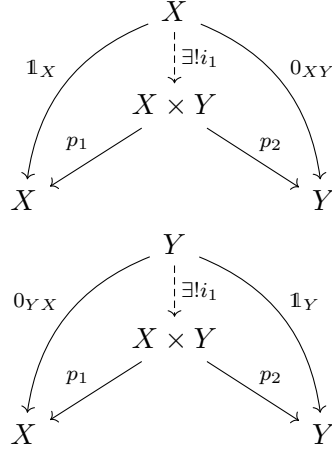
and we see that $h([(a_k)]) = f_{j_k}(a_k)$ as defined above, therefore $h = f$.

Problem 5. Let $\mathcal{C}$ be a pre-additive category and $X, Y \in \mathcal{C}$. Show that if a product $X \times Y$ of X and Y exists, then $X \times Y$ is also a coproduct of X and Y .

Solution: A product $X \times Y$ comes with projections $p_1 : X \times Y \rightarrow X$ and $p_2 : X \times Y \rightarrow Y$ such that the following universal property is satisfied.



So we define $i_1 : X \rightarrow X \times Y$ and $i_2 : Y \rightarrow X \times Y$ using this universal property:



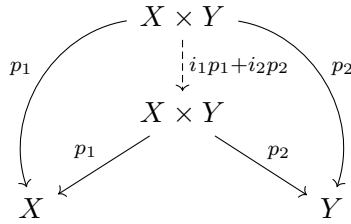
So we have the following properties

$$\begin{cases} \mathbb{1}_X = p_1 i_1, & p_2 i_1 = 0_{XY} \\ \mathbb{1}_Y = p_2 i_2, & p_1 i_2 = 0_{YX} \end{cases} \quad (1.10)$$

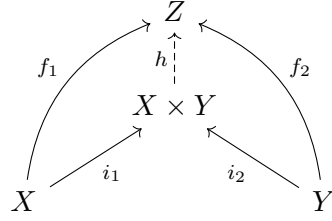
Now, we claim that

$$i_1 p_1 + i_2 p_2 = \mathbb{1}_{X \times Y} \quad (*) \quad (1.11)$$

which can be seen by the diagram below and the universal property of the product.



Now, suppose that we have morphisms $f_1 : X \rightarrow Z, f_2 : Y \rightarrow Z$. Then if we define $h = f_1 p_1 + f_2 p_2$, the diagram below commutes.

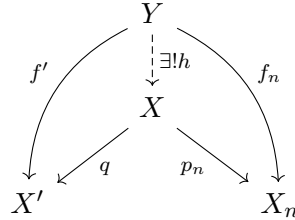


Furthermore, relations (1.10) and equation (*) guarantees that this is the unique expression possible for h that makes the above diagram commutes.

Problem 6. Show that existence of binary products implies existence of all finite n -ary products for $n \geq 2$.

Solution: By a finite n -ary product ($n \geq 2$), we mean an object $X = \prod_{i=1}^n X_i$ with morphisms $p_i : X \rightarrow X_i$ satisfying the following universal property. Take a collection of morphisms $f_i : Y \rightarrow X_i$, ($i = 1, \dots, n$), then there exist an unique morphism $f : Y \rightarrow X$ such that $p_i f = f_i$, $\forall i$. By induction, assume that there exist all finite $(n-1)$ -ary products and let $X' = \prod_{i=1}^{n-1} X_i$. We want to show that $X = X' \times X_n$ satisfies the universal property for a n -ary product. The projections are obtained as follows: we have the projection $p_n : X \rightarrow X_n$ and for $i < n$ we compose the projection $q : X \rightarrow X'$ with the i th projection $p'_i : X' \rightarrow X_i$ to get $p_i : X \rightarrow X_i$.

Now assume we have morphisms $f_i : Y \rightarrow X_i$, ($i = 1, \dots, n$). Then, by the universal property of the finite $(n-1)$ -ary product, there exists a (unique) morphism $f' : Y \rightarrow X'$ such that $p'_i f' = f_i$, $i < n$. Now we have the diagram:



Now we observe that

$$p_i h = (p'_i q) h = p'_i (q h) = p'_i f' = f_i, \quad i < n. \quad (1.12)$$

This show existence of h . Now, for uniqueness, suppose that h' is such that $p_i h' = f_i$, $\forall i$. Then, for $i < n$ we have $p'_i (h' q) = f_i$. So by the universal property of the finite $(n-1)$ -ary product X' , we must have $h' q = f'$. Then, by applying the universal property of the product $X = X' \times X_n$, we must have $h' = h$.

Problem 7. Consider a composition

$$A \longrightarrow B \longrightarrow C$$

in an abelian category, in which $g \circ f = 0$. Using only universal properties, show that there is a canonical morphism from $\text{im}(f)$ to $\text{ker}(g)$. [Recall $\text{im}(f) = \text{ker}(\text{coker}(f))$ by definition.]

Solution: From the universal property of the cokernel we have the diagram.

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & & \\ \downarrow & & \swarrow p & & \downarrow g \\ & & \text{Coker}(f) & \xrightarrow{\exists! \tilde{g}} & C \\ \downarrow & \nearrow & & & \downarrow \\ 0 & \xrightarrow{\quad} & C & & \end{array}$$

Now, by the universal property of the kernel, we have:

$$\begin{array}{ccccccc} A & \xrightarrow{f} & B & & & & \\ \downarrow & \searrow \exists! \tilde{f} & \nearrow i & & g & & \\ & & \text{Ker}(p) = \text{Im}(f) & & & & C \\ \downarrow & \nearrow & \downarrow p & & \nearrow \tilde{g} & & \\ 0 & \xrightarrow{\quad} & \text{coker}(f) & & & & \end{array}$$

Finally, by applying the universal property of the kernel again, we have:

$$\begin{array}{ccccc} \text{Im}(f) & \xrightarrow{i} & B & & \\ \downarrow & \searrow \exists! \gamma & \nearrow & & \downarrow g \\ & & \text{Ker}(g) & & C \\ \downarrow & \nearrow & \downarrow & & \downarrow \\ 0 & \xrightarrow{\quad} & C & & \end{array}$$

as desired.

Problem 8. Let $G : \mathbf{Ab} \rightarrow \mathbf{Set}$ be the forgetful functor, and let $F : \mathbf{Set} \rightarrow \mathbf{Ab}$ be the functor of forming a free abelian group. Show that F is left adjoint to G .

Solution: We have the map

$$\text{Hom}_{\mathbf{Ab}}(F(X), Y) \rightarrow \text{Hom}_{\mathbf{Set}}(X, G(Y)) : f \mapsto f|_X \quad (1.13)$$

Here, we let $F(X) = \{f : X \rightarrow \mathbb{Z} : f(x) \neq 0 \text{ for a finite number of } x \in X\}$ and denote $\hat{x} \in F(X)$ for the map defined by $\hat{x}(y) = \delta_{xy}$. So we see that the set theoretic map $\iota : X \rightarrow F(X) : x \rightarrow \hat{x}$ is an injection and this justifies the notation $f|_X$ used in (*). Moreover, show that this map is an isomorphism is equivalent to show that we have the following universal property: For any set map of sets $f : X \rightarrow G(Y)$, there exists a unique morphism $\hat{f} : F(X) \rightarrow Y$ such that $\hat{f}\iota = f$. So we have no choice but define $\hat{f}(\hat{x}) = f(x)$ and extend to $F(X)$ by $\mathbb{Z}$ -linearity, this defines $\hat{f}$ uniquely.

Problem 9. Compute $(\mathbb{Z}/5) \otimes_{\mathbb{Z}} (\mathbb{Z}/6)$ and $(\mathbb{Z}/p^2) \otimes_{\mathbb{Z}} (\mathbb{Z}/p)$ where p is a prime.

Solution: First we claim that $(\mathbb{Z}/5) \otimes_{\mathbb{Z}} (\mathbb{Z}/6) = (0)$. In fact, if $a \otimes b \in (\mathbb{Z}/5) \otimes_{\mathbb{Z}} (\mathbb{Z}/6)$, then since $6^2 \equiv 1 \pmod{5}$, we have

$$a \otimes b = a \cdot 6^2 \otimes b = a \cdot 6 \otimes 6 \cdot b = a \cdot 6 \otimes 0 = 0. \quad (1.14)$$

For the second group, we claim that $(\mathbb{Z}/p^2) \otimes_{\mathbb{Z}} (\mathbb{Z}/p) \cong \mathbb{Z}/p$. To show this, we show that $\mathbb{Z}/p$ satisfies the universal property of the tensor product (here we will use the notation of problem 10). Let $\varphi_0 : \mathbb{Z}/p^2 \times \mathbb{Z}/p \rightarrow \mathbb{Z}/p : ([m], [n]) \mapsto [mn]$, it's a well defined balanced map, i.e., $\varphi_0 \in \text{Bal}_{A,B}(\mathbb{Z}/p)$. In particular $\mathbb{Z}/p$ is generated by the image of φ_0 . Now, let $f \in \text{Bal}_{A,B}(X)$, we want to define $\tilde{f}$ in such a way the following diagram commutes.

$$\begin{array}{ccc} \mathbb{Z}/p^2 \times \mathbb{Z}/p & \xrightarrow{\varphi_0} & \mathbb{Z}/p \\ f \downarrow & \swarrow \tilde{f} & \\ X & & \end{array}$$

So we set $\tilde{f}([n]) = f(1, [n])$, which is well defined while makes the diagram commutes.

Problem 10. Let R be a unital ring. In order to formulate the definition of tensor product of R -modules, we introduce a definition:

definition 0.1. Let A be a right R -module and B a left R -module, and X an abelian group. A balanced map $f \in \text{Bal}_{A,B}(X)$ is a function $f : A \times B \rightarrow X$ such that

$$f(a_1 + a_2, b) = f(a_1, b) + f(a_2, b) \quad \text{for all } a_1, a_2 \in A \text{ and } b \in B$$

$$f(a, b_1 + b_2) = f(a, b_1) + f(a, b_2) \quad \text{for all } a \in A \text{ and } b_1, b_2 \in B$$

$$f(ar, b) = f(a, rb) \quad \text{for all } a \in A, b \in B \text{ and } r \in R.$$

Now a tensor product of a right R -module A with a left R -module B is an abelian group $A \otimes_R B$ together with a $\varphi_0 \in \text{Bal}_{A,B}(A \otimes_R B)$, such that for any balanced map f as below, there exists a unique linear map $\tilde{f}$ making the diagram commute.

$$\begin{array}{ccc} A \times B & \xrightarrow{\varphi_0} & A \otimes_R B \\ & \searrow f & \swarrow \tilde{f} \\ & X & \end{array}$$

Verify that

$$A \otimes_R B \cong Q = \frac{A \otimes_{\mathbb{Z}} B}{\langle ar \otimes b - a \otimes rb \mid a \in A, b \in B, r \in R \rangle}.$$

Solution: To prove the asserted isomorphism, we show that the quotient of abelian groups Q satisfies the universal property of the tensor product $A \otimes_R B$. Let $f \in \text{Bal}_{A,B}(X)$, then f is balanced by viewing A, B as abelian groups. By the universal property of $A \otimes_{\mathbb{Z}} B$, there exist maps $\hat{\varphi}_0$ and $\hat{f}$, such that:

$$\begin{array}{ccc} A \times B & \xrightarrow{\hat{\varphi}_0} & A \otimes_{\mathbb{Z}} B \\ & \searrow f & \swarrow \hat{f} \\ & X & \end{array}$$

Now, since $f \in \text{Bal}_{A,B}(X)$, we have

$$\text{Ker}(\hat{f}) \supset \langle ar \otimes b - a \otimes rb \mid a \in A, b \in B, r \in R \rangle. \quad (1.15)$$

Then, there exists $\tilde{f}$ such that the following diagram commutes:

$$\begin{array}{ccccc} A \times B & \xrightarrow{\hat{\varphi}_0} & A \otimes_{\mathbb{Z}} B & \xrightarrow{p} & Q \\ & \searrow f & \swarrow \hat{f} & & \uparrow \tilde{f} \\ & X & & & \end{array}$$

Now, by construction, the quotient Q is generated by the image of the map $\varphi_0 = p\hat{\varphi}_0$ and since $f = \tilde{f}\varphi_0$ we see that $\tilde{f}$ is uniquely determined by f . Therefore (Q, φ_0) satisfies the universal property for the tensor product $A \otimes_R B$ and $Q \cong A \otimes_R B$.